

Auteur: Seda Jusjajeva
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$$f(x, y) = 4xy - 2x^4 - y^2$$

$$\begin{cases} \frac{\partial f}{\partial x}(x, y) = 4y - 8x^3 \\ \frac{\partial f}{\partial y}(x, y) = 4x - 2y \end{cases}$$

$$\begin{cases} -8x^3 + 4y = 0 \\ 4x - 2y = 0 \end{cases}$$

$$\Leftrightarrow \begin{cases} -8x^3 + 8x = 0 \\ y = 2x \end{cases}$$

$$\Leftrightarrow \begin{cases} 8x(-x^2 + 1) = 0 \\ y = 2x \end{cases}$$

$$\Leftrightarrow \begin{cases} 8x = 0 \vee -x^2 + 1 = 0 \\ y = 2x \end{cases}$$

$$\Leftrightarrow \begin{cases} x = 0 \vee x = -1 \vee x = 1 \\ y = 0 \vee y = -2 \vee y = 2 \end{cases}$$

$$\nabla f(x, y) = 0 \text{ als } (x, y) = (-1, -2) \vee (x, y) = (0, 0) \vee (x, y) = (1, 2)$$

Er zijn dus 3 kritieke punten $(-1, -2)$, $(0, 0)$ en $(1, 2)$

$$\Delta(x, y) = \begin{vmatrix} -24x^2 & 4 \\ 4 & -2 \end{vmatrix} = 48x^2 - 16$$

$$\Delta(-1, -2) = \Delta(1, 2) = 48 - 16 = 32 > 0 \Rightarrow \text{minima/maxima}$$

$$(A = -24x^2)$$

$$A(-1, -2) = A(1, 2) = -24 \cdot 1 = -24 < 0 \Rightarrow \text{maxima}$$

$$\Delta(0, 0) = -16 < 0 \Rightarrow \text{zadelpunt}$$

References